

CO4 AT2 CLOUD:

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1. Retail Data Optimization

Architecture:

POS + Online Store + Customer Data → Kafka → Data Lake → Spark → ML Models
→ Dashboard/Recommendation System

- **Data sources:** Store sales, online orders, inventory, customer clicks, browsing and purchase history.
 - **Data ingestion:** **Apache Kafka** collects real-time data from stores and the online platform.
 - **Storage:** **Hadoop HDFS / Cloud Data Lake** stores large historical datasets.
 - **Processing:** **Apache Spark** performs batch and real-time processing.
 - **Analytics:**
 - Demand forecasting → **Time-series models, Random Forest, LSTM**
 - Customer segmentation → **K-Means clustering**
 - Personalized recommendations → **Collaborative filtering / recommendation systems**
 - **Output:** Real-time inventory dashboard, automatic stock alerts and personalized product recommendations.
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2. Social Media Fraud Detection

Architecture:

Posts + Likes + Comments + Shares → Kafka → Spark Streaming/Flink → ML Models → Fraud Detection System

- **Data ingestion:** **Apache Kafka** collects billions of interactions continuously.
- **Stream processing:** **Apache Spark Streaming or Apache Flink** processes events in real time.
- **Storage:** **Hadoop/HDFS or cloud data lake** stores historical interactions.
- **Features:** Account age, posting frequency, IP/device patterns, follower relationships, content similarity and interaction behavior.

- **Machine Learning:**
 - Fake-account detection → **Random Forest, XGBoost, Neural Networks**
 - Misinformation detection → **NLP, text classification, transformer-based models**
 - Suspicious networks → **Graph analytics**
 - **Output:** Suspicious accounts/content are given a risk score and flagged for review.
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3. Government Data Platform

Architecture:

Census + Economic + Geospatial Data → Data Integration → Data Lake/Warehouse
→ Governance Layer → Departments

- **Data storage:** A centralized **data lake** stores structured and unstructured government data.
- **Integration:** **ETL/ELT tools** integrate data from different departments.
- **Interoperability:** Common data formats, schemas, APIs and metadata standards are used.
- **Data governance:**
 - Data ownership and stewardship
 - Data quality management
 - Metadata and data catalog
 - Standardized data definitions
- **Security:**
 - Role-Based Access Control (RBAC)
 - Encryption
 - Authentication and authorization
 - Audit logs
 - Data masking/anonymization
- **Geospatial analytics:** GIS tools process location-based datasets.
- **Result:** Departments can securely share and access consistent national data.

4. Banking Fraud Detection

Architecture:

Transactions → Kafka → Stream Processing → Feature Engineering → ML Model → Fraud Alert

- **Data sources:** ATM, UPI, cards, online banking and historical transactions.
- **Real-time ingestion:** **Apache Kafka** handles high-volume transaction streams.
- **Processing:** **Apache Flink / Spark Streaming** analyzes transactions with low latency.
- **Storage:** Data lake/Hadoop stores historical transaction data.
- **Machine Learning:**
 - **Random Forest / XGBoost** → fraud classification
 - **Isolation Forest** → anomaly detection
 - **Neural Networks** → complex transaction patterns
- **Features:** Transaction amount, location, time, device, frequency and customer behavior.
- **Decision:** Model generates a fraud probability score.
- **Action:** High-risk transactions are blocked or sent for immediate verification.

5. Agriculture Smart Farming

Architecture:

Sensors + Drones + Weather Stations → IoT Gateway → Kafka → Data Lake → Spark/ML → Farmer Dashboard

- **Data sources:** Soil moisture, temperature, humidity, weather, crop images and drone data.
- **Data ingestion:** IoT gateways collect sensor data; **Kafka** handles continuous data streams.
- **Storage:** Cloud data lake/HDFS stores historical sensor and image data.
- **Processing:** **Apache Spark** analyzes large-scale agricultural data.
- **Analytics:**
 - Crop health → **Image processing + CNN**

- Yield prediction → **Random Forest / Regression**
- Irrigation prediction → **ML models based on soil moisture and weather**
- Disease detection → **CNN/deep learning**
- **Decision support:** Farmers receive alerts for irrigation, fertilizer application, crop diseases and weather risks.
- **Result:** Improved crop yield, reduced water/fertilizer usage and better resource management.